

## Tianyi Chen

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### EDUCATION

**Johns Hopkins University** *Baltimore, Maryland*  
PhD candidate at Applied Math and Statistics Department Sep 2021- May 2026 (Expected)

**Johns Hopkins University** *Baltimore, Maryland*  
Master of Science in Engineering Statistics Major Aug 2019 - May 2021

**Renmin University of China** *Beijing, China*  
Bachelor of Science Statistics Major Economics Statistics Minor Sep 2015 - June 2019

### PUBLICATIONS AND PREPRINTS

Helm, H., **Chen, T.**, McGuinness, H., Lee, P., Duderstadt, B., & Priebe, C. E. (2025). Toward a digital twin of US Congress. *arXiv:2505.00006*.

**Chen, T.**, Lubberts, Z., Athreya, A., Park, Y., & Priebe, C. E. (2024). Euclidean mirrors and first-order changepoints in network time series. *arXiv:2405.11111*. Major revision at Electronic Journal of Statistics.

**Chen, T.**, Park, Y., Saad-Eldin, A. et al. Discovering a change point and piecewise linear structure in a time series of organoid networks via the iso-mirror. *Applied Network Science* 8, 45 (2023).

Priebe, C. E., Shen, C., Huang, N., & **Chen, T.** (2021). A Simple Spectral Failure Mode for Graph Convolutional Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Wang, F., **Chen, T.**, Chang, Q., Kao, Y. W., Li, J., Chen, M., ... & Shia, B. C. (2021). Respiratory diseases are positively associated with PM2. 5 concentrations in different areas of Taiwan. *Plos one*, 16(4), e0249694.

**Chen, T.**, Chen, Y., Gao, J., Gao, P., Moon, J. H., Ren, J., ... & Woolf, T. B. (2021). Machine Learning to Summarize and Provide Context for Sleep and Eating Schedules. *bioRxiv*, 2020-12.

### RESEARCH IN PROGRESS

**Human Brain Organoid Connectome Analysis** May 2023 – Present  
*Collaboration with JHU Center for Alternatives to Animal Testing and Applied Physics Laboratory*

- Developed a pipeline that processes time series of brain organoid Multi-Electrode Array (MEA) recording data into Euclidean mirror representations. Explored the connectome's dynamic evolution over time.

### TEACHING EXPERIENCE

Teaching Assistant at *Johns Hopkins University Applied Mathematics and Statistics Department*

- Course “Statistical Pattern Recognition Theory & Methods” Spring 2025
- Course “Statistics Theory” Fall 2021 Fall 2024
- Course “Time series analysis” Spring 2024

- Course “Intro to Optimization” Spring 2020
- Course “Applied Statistics and Data Analysis” Fall 2020 Fall 2022 Fall 2023
- Course “Applied Statistics and Data Analysis 2” Spring 2021

Awarded Teaching Fellow

May 2025

## TALKS

“Vertex alignment and first-order changepoint in time series of graphs”

*Joint Statistical Meeting, Nashville*

Aug 4 2025

“Toward a digital twin of US Congress.”

*Symposium on Data Science and Statistics, Salt Lake City*

April 30 2025

“Vertex alignment and first-order changepoint in time series of graphs”

*JHU Applied Math and Statistics department Student Seminar*

March 25 2025

“Euclidean mirror and first-order changepoint in network time series”

*SIAM Conference on Mathematics of Data Science, Poster session, Atlanta*

Oct 23 2024

“Euclidean mirror and first-order changepoint in network time series”

*38<sup>th</sup> New England Statistics Symposium, invited talk, University of Connecticut*

May 20 2024

“Organoid intelligence and time series of networks.”

*JHU Applied Math and Statistics department Student Seminar*

Sep 26 2023

“A simple example for understanding swarms.”

*JHU Applied Math and Statistics department Student Seminar*

Oct 11 2022

“Understanding swarm: manifold learning on spectral representation of time series of graphs.”

*JHU Applied Math and Statistics department Student Seminar*

Oct 19 2021

## AWARDS AND FELLOWSHIPS

Nominated as Seibel scholar by Applied Mathematics and Statistics Department

June 2025

IMSI “Statistics Meets Tensors” workshop travel fund

May 2025

Honorable Mention Audience Choice Award at the DSAI End of Semester Social for presentation of “Toward a digital twin of US Congress.”

May 2025

Teaching Fellow Recognition Award

May 2025

Duncan Fund for travel

May 2022, May 2024, Aug 2024, Oct 2024, May 2025

MINDS (Mathematical Institute of Data Science) Fellowship

Spring 2022

## SOCIAL ACTIVITIES

Served as AMS diversity committee

Sep 2024 - May 2025

Organized AMS special seminar: Converging Perspectives

April 4 2025

Helped organize and attended 7th and 9th International Forum on Statistics

May 2016 - Jul 2018